



Transport-Based Joint Distribution Alignment for Multi-site Autism Spectrum Disorder Diagnosis Using Resting-State fMRI

Junyi Zhang, Peng Wan, and Daoqiang Zhang^(✉)

College of Computer Science and Technology,
Nanjing University of Aeronautics and Astronautics, Nanjing, China
dqzhang@nuaa.edu.cn

Abstract. Resting-state functional magnetic resonance imaging (rs-fMRI) has been a promising technique for computer-aided diagnosis of neurodevelopmental brain diseases, *e.g.*, autism spectrum disorder (ASD), due to its sensitivity to the progressive variations of brain functional connectivity. To overcome the challenge of overfitting resulting from small sample size, recent studies began to fuse multi-site datasets for improving model generalization. However, these existing methods generally simply combine multiple sites into single dataset, ignoring the heterogeneity (*i.e.*, data distribution discrepancy) among diverse sources. Actually, *distribution alignment*, alleviating the inter-site distribution shift, is the fundamental step for multi-site data analysis. In this paper, we propose a novel Transport-based Multi-site Joint Distribution Adaptation (TMJDA) framework to reduce multi-site heterogeneity for ASD diagnosis by aligning joint feature and label distribution using optimal transport (OT). Specifically, with the given target domain and multi-source domains, our TMJDA method concurrently performs joint distribution alignment in each pair of source and target domains, upon which, multiple domain-specific classifiers are further aligned by penalizing decision inconsistency among diverse classifiers for reducing inter-site distribution discrepancy. We evaluated our TMJDA model on the public Autism Brain Imaging Data Exchange (ABIDE) database. Experimental results demonstrate the effectiveness of our method in ASD diagnosis based on multi-site rs-fMRI datasets.

Keywords: Multi-site · Optimal transport · Autism spectrum disorder

1 Introduction

Autism spectrum disorder (ASD), characterized by deficits in social communication and the presence of restricted, repetitive behaviors [1], is the most

J. Zhang and P. Wan—These authors contributed equally to this work.

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prevalent neurodevelopmental disorder that ultimately seriously affects individual social and daily life. Considerable studies [2, 3] have proven that the variations of brain functional connectivity pattern (*e.g.*, local overconnectivity and long-distant underconnectivity) are associated with ASD and its progression, especially considering that this changing process occurs even earlier than the appearance of disconnection syndromes. Resting-state fMRI can non-invasively capture profound brain connectivity variations induced by the pathological process. Therefore, various learning-based methods [4–6] have been proposed for automatically mining the relevant local or global connectivity characteristics which are used as important biomarkers in ASD diagnosis.

Nonetheless, clinical practical applications of these studies are still faraway due to the limitation of reproducibility across diverse autism research centers [7]. This mainly arises from the potential variations across individuals, MR scanners, and data acquisition protocols [8]. In terms of statistical learning, the resulting data heterogeneity, *i.e.*, distribution discrepancy, severely degrades the model generalization when applied to a new target imaging domain. In recent years, a number of techniques [9–11] have been developed to deal with the problem of distribution shift across multiple data sites, typically Unsupervised Domain Adaptation (UDA), which leverages knowledge learned from one or more related source domains to facilitate classifier construction in target domain with no annotated samples [12]. One of the common strategies is to seek a transformation $\mathcal{T}$ such that the new representations of input data are matching (*i.e.*, $P_s(\mathcal{T}(X)) = P_t(\mathcal{T}(X))$) under some distribution measures (*e.g.*, Bergman Divergence [13], Maximum Mean Discrepancy (MMD) [14] and et al. [15, 16]). A particular case is the use of optimal transport (OT) [17, 18], which seeks for an optimal transport plan γ by minimizing the global transportation cost between the empirical distributions of X_s and X_t , thereby interpolating X_s via barycentric mapping and ultimately bringing the source and target distribution closer. In this way, a shared embedding space could be learned while preserving the discriminative information [19], which is suitable to our task since the goal is to mine the common brain connectivity variation pattern across multiple ASD research centers.

In this paper, we propose a novel multi-source unsupervised domain adaptation framework using optimal transport, named as Transport-based Multi-site Joint Distribution Adaptation (TMJDA), to reduce the heterogeneity of multi-site resting-state fMRI data for improving ASD diagnostic performance. As shown in Fig. 1, our proposed TMJDA model admits a multiway parallel adaptation framework. To be specific, each source domain is adapted to the target domain using our proposed transport-based joint distribution alignment block (T-JDA block), meanwhile, a multi-classifier inconsistency cost is incorporated for reducing the mismatching of outputs of different classifiers, under the circumstance of knowledge propagation among diverse sources. The main innovations of our proposed TMJDA model can be summarized as follows, (1) each T-JDA block can propagate *feature/label* from source to target by minimizing the global transportation cost between empirical joint distribution of two

domains; (2) the parallel adaptation architecture guided by an inconsistency loss can further propagate knowledge transfer across all data sites.

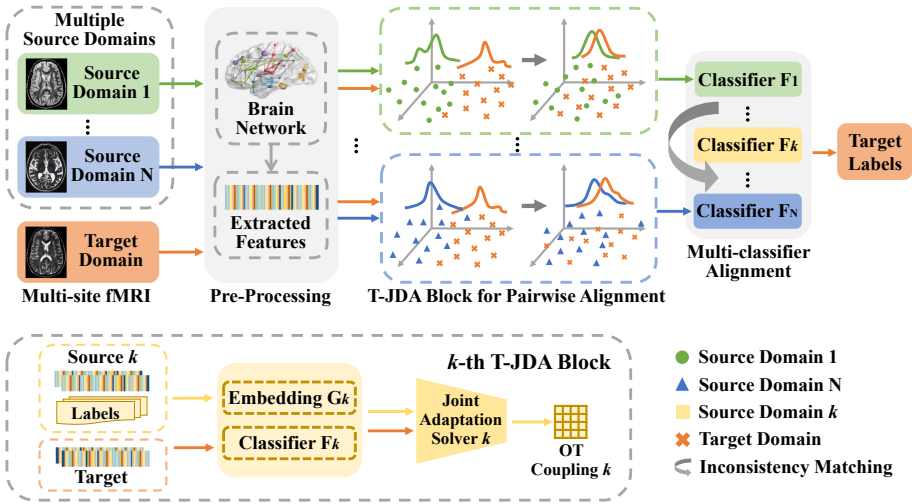


Fig. 1. An overview of the proposed Transport-based Multi-site Joint Distribution Adaptation (TMJDA) framework. Our framework receives labeled multi-source samples and a target domain with no annotated samples. Specific T-JDA block performs joint distribution alignment in each pair of source and target domains while a shared classifier F is learnt in both domains, upon which, multiple classifiers are aligned to classify target samples. (Best viewed in color)

2 Materials and Method

2.1 Datasets and Image Pre-processing

The Autism Brain Imaging Data Exchange [20] database is a qualified multi-site heterogeneous data collection. It contains resting-state fMRI and clinical data on altogether 1,112 subjects from 17 diverse sites. Considering that several sites suffer from a small sample size, five sites with more than 50 subjects contributes to our work: *Leuven*, *USM*, *UCLA*, *UM* and *NYU*. These sites contain 468 subjects, including 218 patients with ASD and 250 age-matched typical control individuals (TCs).

Considering the reproducibility of our work, we used the published version of this database in Preprocessed Connectome Project initiative. Firstly, Configurable Pipeline for the Analysis of Connectomes (C-PAC) [21] was used to preprocess resting-state fMRI, including slice-timing correction, image realignment and nuisance regression. After that, time series of average voxel signals were taken from each brain region extracted by the predefined Anatomical Automatic Labeling (AAL) atlas with 116 ROIs. Based on that, the correlation between each

pair of ROIs is estimated using Pearson Correlation Coefficient, resulting a 116×116 symmetrical matrix representing functional connectivity for each subject. Finally, we utilized the lower-triangular matrix without 116 diagonal elements and flatten it to 6,670-dimensional features for each subject.

2.2 Transport-Based Multi-site Joint Adaptation

The standard learning paradigm of multi-source unsupervised domain adaptation (MUDA) assumes the existence of N source datasets $\{(X_s^k, Y_s^k)\}_{k=1}^N$, $Y_s^k = \{y_i^{sk}\}_{i=1}^{n_{sk}}$ is the associated class label for k -th source domain, and a target set of data $X_t = \{x_j^t\}_{j=1}^{n_t}$ with unknown labels. In the context of UDA, we assume that these joint probability distributions of source domain $P_s^k(X, Y)$ differ from that of target domain $P_t(X, Y)$. In order to determine the set of labels Y_t associated with X_t , the core task is to align empirical joint distribution $\hat{P}_s^k(X, Y)$ and $\hat{P}_t(X, Y)$, with expect to transfer knowledge from multiple sources to assist decision boundary learning in target domain.

Overall Adaptation Framework. Figure 1 shows a schematic diagram of our Transport-based Multi-site Joint Distribution Adaptation (TMJDA) framework, characterized by a multiway parallel adaptation architecture, which consists of 1) data preprocessing block, 2) transport-based joint distribution alignment block (T-JDA block) and 3) multi-classifier alignment. After data preprocessing, T-JDA block performs joint distribution alignment in each pair of source and target domains in parallel, accompanied by multi-classifier inconsistency loss to propagate knowledge transfer across all domains.

Transport-Based JDA Block. Given one pair of source and target domain, traditional assumption is that, the conditional distributions are kept unchanged under transformation $\mathcal{T}$, *i.e.*, $P_s(Y|\mathcal{T}(X)) = P_t(Y|\mathcal{T}(X))$, but it remains controversial in practical problems. A more reasonable assumption should admit there exist shift in both marginal and conditional probabilities, that is, one should permit the direct alignment of joint distributions $P_s(X, Y)$ and $P_t(X, Y)$. Following the Kantorovich formulation [22], Courty et al. [23] proposed to seek a probabilistic coupling γ between empirical joint distributions, *i.e.*, $\hat{P}_s = \frac{1}{n_s} \sum_{i=1}^{n_s} \delta_{x_i^s, y_i^s}$ and $\hat{P}_t = \frac{1}{n_t} \sum_{j=1}^{n_t} \delta_{x_j^t, y_j^t}$, which minimizes the transport cost:

$$\gamma_0 = \arg \min_{\gamma \in \Pi(\hat{P}_s, \hat{P}_t)} \sum_{ij} D(x_i^s, y_i^s; x_j^t, y_j^t) \gamma_{ij} \quad (1)$$

where $\Pi = \{\gamma \in (\mathbf{R}^+)^{n_s \times n_t} | \gamma \mathbf{1}_{n_s} = \hat{P}_s, \gamma^T \mathbf{1}_{n_t} = \hat{P}_t\}$ and $\mathbf{1}_d$ is a d -dimensional one vector. The joint cost $D(x_i^s, y_i^s; x_j^t, y_j^t) = \alpha c(x_i^s, x_j^t) + \mathcal{L}(y_i^s, y_j^t)$ is a weighted combination of the distances in feature space and the classification loss $\mathcal{L}$ (*e.g.*, hinge or cross-entropy) in label space. Since no label is available in the target domain, y_j^t is replaced by a proxy $f(x_j^t)$ using predictive function f .

Inspired by this, we propose a basic T-JDA block to learn a shared common space which shrinks the feature/label discrepancy between domains. The block consists of three components, *i.e.*, 1) shared feature embedding function $g_k(\cdot)$, 2) domain-specific classifier $f_k(\cdot)$ and 3) joint distribution adaptation solver $\mathcal{L}_{t-jda}^k$. Upon which, T-JDA block jointly aligns the feature space and label conditional distribution, such that the estimation of both mapping $g_k(\cdot)$ and classifier $f_k(\cdot)$ is embedded in the optimization of optimal transport coupling γ between $\hat{P}_s^k(X, Y)$ and $\hat{P}_t(X, Y)$.

We use Eq. (1) as the estimate of adaptation cost in each T-JDA block, where $c(\cdot, \cdot)$ is chosen as the squared Euclidean metric and $\mathcal{L}_t$ is a traditional cross-entropy loss. The T-JDA solver is reformulated as:

$$\mathcal{L}_{t-jda}^k = \sum_{i=1}^{n_{sk}} \sum_{j=1}^{n_t} \gamma_{ij}^k (\alpha \|g_k(x_i^{sk}) - g_k(x_j^t)\|^2 + \mathcal{L}_t(y_i^{sk}, f_k(g_k(x_j^t)))) \quad (2)$$

Multi-classifier Inconsistency Matching. Since domain-specific classifiers in each T-JDA block are learned from diverse source-target domain pairs, the outputs of these classifiers may be inconsistent when predicting the same target sample, especially for samples near domain-specific decision boundary [24]. In order to reduce the prediction discrepancy among multiple classifiers while facilitate knowledge propagation among multiple diverse sources, we penalize decision inconsistency by minimizing the absolute value of the discrepancy in the outputs of all pairs of classifiers (*i.e.*, $f_k(\cdot)$ and $f_m(\cdot)$) when classifying target samples. The inconsistency loss is formulated as:

$$\mathcal{L}_{inc} = \frac{1}{N \times (N-1)} \sum_{k,m=1}^N \mathbb{E}_{x \sim X_t} [|f_k(g_k(x_j^t)) - f_m(g_m(x_j^t))|] \quad (3)$$

Multi-site Joint Alignment and Classification. Our TMJDA model performs joint distribution alignment and ASD diagnostic based on multi-site rs-fMRI datasets. It optimizes the learnable parameters for each basic T-JDA block in parallel, while updating the classifier both in source and target domains. It is worth noting that, the mapping $g_k(\cdot)$ and classifier $f_k(\cdot)$ are shared by source-target domain pairs in k -th T-JDA block. As shown in Fig. 1, the total loss function $\mathcal{L}_{total}$ is a combination of adaptation cost in multiway T-JDA block (*i.e.*, $\mathcal{L}_{t-jda}$), source-domain classification loss (*i.e.*, $\mathcal{L}_{cls}$) and multi-classifier inconsistency cost in target domain (*i.e.*, $\mathcal{L}_{inc}$). To be specific, we use Eq. (2) as the adaptation cost in each T-JDA block, upon which, the multiway adaptation cost is reformulated as $\mathcal{L}_{t-jda} = \sum_{k=1}^N \mathcal{L}_{t-jda}^k$. Considering the possible performance degradation of classifier in source domain, *i.e.*, catastrophic forgetting [25], due to overemphasis on features and labels alignment, a classification loss using cross-entropy is incorporated here, which is formulated as:

$$\mathcal{L}_{cls} = \sum_{k=1}^N \mathbb{E}_{x \sim X_s^k} \mathcal{L}_s(y_i^{sk}, f_k(g_k(x_i^{sk}))) \quad (4)$$

We define total loss function as:

$$\mathcal{L}_{total} = \mathcal{L}_{cls} + \mathcal{L}_{t-jda} + \lambda \mathcal{L}_{inc} \quad (5)$$

where λ is a parameter to control the contribution of multi-classifier inconsistency cost. Finally, we average the outputs of all domain-specific classifiers as the prediction of target sample by majority voting.

Optimization. The loss function is optimized by an alternative minimization approach [26]. We start with submodule (g_k, f_k) pre-trained on the corresponding labeled source domain. At the training stage, we randomly sample a batch of n_b samples from one of N source domains and a batch of n_b samples from target domain, and perform optimization as follows:

(1) With fixed parameters of submodule $(\hat{g}_k, \hat{f}_k)$, the optimization of T-JDA solver is a standard OT problem, and we can obtain coupling γ_k by minimizing the transport cost:

$$\min_{\gamma_k \in \Pi(\hat{P}_s, \hat{P}_t)} \sum_{i,j=1}^{n_b} \gamma_{ij}^k (\alpha \|\hat{g}_k(x_i^{sk}) - \hat{g}_k(x_j^t)\|^2 + \mathcal{L}_t(y_i^{sk}, \hat{f}_k(\hat{g}_k(x_j^t)))) \quad (6)$$

(2) With fixed couplings $\hat{\gamma}_1 \dots \hat{\gamma}_N$ obtained at previous step, we concurrently update multiple submodules $(g_1, f_1) \dots (g_N, f_N)$, which follows classical mini-batch stochastic gradient descent (SGD) algorithm. We can obtain submodule (g_k, f_k) by minimizing the total loss:

$$\begin{aligned} & \frac{1}{n_b} \sum_{i=1}^{n_b} \mathcal{L}_s(y_i^{sk}, f_k(g_k(x_i^{sk}))) + \sum_{i,j=1}^{n_b} \hat{\gamma}_{ij}^k (\alpha \|g_k(x_i^{sk}) - g_k(x_j^t)\|^2 \\ & + \mathcal{L}_t(y_i^{sk}, f_k(g_k(x_j^t)))) + \lambda \frac{1}{N-1} \sum_{m=1}^N \frac{1}{n_b} \sum_{j=1}^{n_b} (|f_k(g_k(x_j^t)) - f_m(g_m(x_j^t))|) \end{aligned} \quad (7)$$

Implementation. Our TMJDA model was implemented using Python based on Keras. For simplicity, feature embedding function consists of only two fully connected layers of 1024 and 128 hidden units followed by a ReLU nonlinearity respectively, and domain-specific classifier consists of a fully connected layer of 64 hidden units follow by a softmax layer. Besides, we set the batch size to 32, learning rate to 0.01 and SGD momentum to 0.01. Instead of fixing the trade-off parameters, *i.e.*, α in $\mathcal{L}_{t-jda}$ and λ of $\mathcal{L}_{inc}$, we set $\alpha_k = 1/\max \|g_k(x_i^{sk}) - g_k(x_j^t)\|^2$ to normalize the range of distances in domain-specific feature space [23], and set $\lambda_p = 1/(1 + \exp(-10p)) - 1$ to suppress noisy activations from multi-classifier at early training phases [27].

3 Experiments

3.1 Experimental Setup

We perform an application for multi-site datasets from the ABIDE database to evaluate the diagnostic performance of Transport-based Multi-site Joint Distribution Adaptation (TMJDA) model for ASDs and TCs. In our experiment, we alternately treat one site as target domain, and the rest are sources. The evaluation metrics including classification accuracy (ACC), sensitivity (SEN) and specificity (SPE). Let us note TP (*resp.*, TN) as true positives (*resp.*, negatives) and FP (*resp.*, FN) as false positives (*resp.*, negatives), we define $ACC = (TP + TN)/(TP + FP + TN + FN)$, $SEN = TP/(TP + FN)$ and $SPE = TN/(TN + FP)$, where ACC corresponds to the ability of model to predict correctly, SEN corresponds to the ability to detect healthy individuals and SPE corresponds to the ability to detect patients with ASD.

Considering there is a small amount of work involved in multi-site adaptation for disease diagnosis, in our experiment, we choose a recent method Multi-center Low-Rank Representation (MCLRR) [9] as multi-site adaptation baseline, and compare TMJDA model with multi-site combine methods, including Denoising Auto-Encoder (DAE) [6] and Optimal Transport Domain Adaptation (OTDA) [19]. Under this setting, we testify whether multi-site heterogeneity has an impact on ASD diagnosis, and demonstrates the effectiveness of our TMJDA model. Furthermore, we evaluate a variant TMJDA-1, *i.e.*, without multi-classifier inconsistency cost, to demonstrate the contribution of parallel domain-specific mapping and multi-classifier alignment to our model.

3.2 Results and Discussion

We compare TMJDA with or without inconsistency cost and baseline methods on five data sites and show the performance comparison of classification accuracy in Fig. 2. Furthermore, we report the detailed results on *UM* site in Table 1. From these results, we get the following insightful observations.

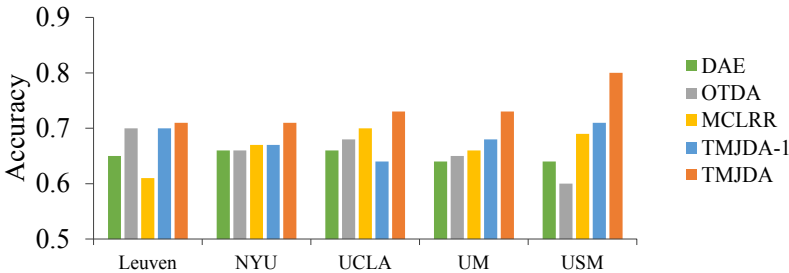


Fig. 2. Performance comparison of classification accuracy for ASDs and TCs on multiple sites from ABIDE database.

In Fig. 2, it shows that TMJDA outperforms all compared baseline methods. The average classification accuracy across multiple sites achieved by these methods are 65.00%, 65.82%, 66.76%, 68.11% and 73.29% respectively, which demonstrates the effectiveness of our method in ASD diagnosis based on multi-site rs-fMRI datasets.

Table 1. Results of classification accuracy, sensitivity and specificity on *UM* site.

	Method	ACC (%)	SEN (%)	SPE (%)
Multi-site combined	DAE [6]	64.00	66.00	62.00
	OTDA [19]	65.49	70.83	61.53
Multi-site adaptation	MCLRR [9]	66.30	75.16	59.53
	TMJDA-1	68.14	66.67	69.23
	TMJDA (ours)	72.57	75.00	70.77

In Table 1, comparing multi-site combine and multi-site adaptation methods, it is obvious that the multi-site heterogeneity can have a negative effect on ASD diagnosis, even the former one benefits from data expansion. TMJDA-1 outperforms MCLRR in most cases, it indicates that compared with the latter one, which tries to map all domains into a common latent space, TMJDA-1 is simpler and more efficient to concurrently performs joint distribution alignment in multiple source-target domain pairs to seek domain-specific mapping. Comparing TMJDA with or without inconsistency cost, TMJDA achieves much better performance than TMJDA-1, which indicates that aligning domain-specific classifiers by penalizing the decision inconsistency among diverse classifiers can furtherly propagate knowledge transfer across diverse domains.

4 Conclusions

In this paper, we propose a novel Transport-based Multi-site Joint Distribution Adaptation (TMJDA) framework to reduce the heterogeneity of multi-site resting-state fMRI data for improving ASD diagnostic performance. Our framework concurrently performs joint distribution alignment using optimal transport in diverse source-target domain pairs, and the parallel adaptation architecture guided by multi-classifier alignment to classify target samples. Experimental results on the public ABIDE database demonstrate the effectiveness of our method. Furthermore, the TMJDA framework holds the promise to be applied to other multi-site databases.

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